Tutorial 9

This week students will revisit the recursion concept and exception handling.

TAs will describe what is recursion. How some problems such as Fibonacci(or Tribonacci from the lab) sequence can be modeled using recursion, i.e, function within a function.

Another example

<u>Geometric progression as recursion</u>

Given a geometric sequence with initial term **a0** and common ratio **r**, how can we find the **nth** term?
One simpler way (and a good one too) is simply raise **r** to the exponent of (n-1) and multiply with **a0.**

```python
##Geometric Progression
a0 = 1
r  = 2
n  = 6

print("Nth term of the GP is:", a0*(2**(n-1)))
```

Output

```
Nth term of the GP is: 32
```

The same can be done recursively,

```
##Geometric Progression
a0 = 1
r  = 2
n  = 6

#print("Nth term of the GP is:", a0*(2**(n-1)))

## let's define a function

def gp_term(n, a0, r):
     ## here 3 parameters have been provided

     if n==1:
          return a0
     else:
          return r*gp_term(n-1, a0, r)


nth_term = gp_term(n, a0, r)

## n, a0 and r have already been defined

print("Nth term of the GP is[using recursion]:", nth_term)
```

Output

```
Nth term of the GP is[using recursion]: 32
```

TAs would ask the students: Can they make this function a generic one, i.e, for any a0, r and n? The students should try on their own.

Exception Handling

TAs will discuss different types of exceptions, discussed by Dr. Pankaj in the class.

Then a program on exception handling would be done. This program also uses file handling, so the students will be able to revisit those concepts as well!

Consider, one tries to read a file that doesn't exist!!!

```
## currently no file named 'example.txt' in current directory

file = open("example.txt", 'r')

##  opening with read  access

file.read()

## what should happen?let's see
```

Output

```
Traceback (most recent call last):
  File "C:\Users\Shashwat\Desktop\IntroProg\Lab 9\tut9 exception handling\except
Handle.py", line 4, in <module>
    file = open("example.txt", 'r')
FileNotFoundError: [Errno 2] No such file or directory: 'example.txt'
>>>
```

We encountered “FileNotFoundError”!!! Exactly as expected!!

But we still do need our program to run without getting interrupted by throwing some error! In that case, we need to handle this smartly, using our try and except blocks…

Suppose whenever the above case happens, the program creates a new file and then reads it. If the file already exists, then also it should have worked nonetheless.

I.e, if suppose a file “abc.txt ” exists but “example.txt” doesn’t.

Note to the TAs: abc.txt contains these two lines

Hello students

Welcome to the tutorial on exception handling

```
files_list = [  "example.txt", "abc.txt"]
##we know that example.txt doesn't exist

for f in files_list:
     try:
          print(f'opening {f}')
          file = open(f, 'r+')
          print(file.read())

     except:
          print(f'{f} doesn\'t exist')
```

Output:

```
opening example.txt
example.txt doesn't exist
opening abc.txt
Hello students
Welcome to the tutorial on exception handling!
```

As can be seen, we avoided the error-induced interruption using exception handling.

At the end of the tutorial, students should have revisited recursion, functions, file handling and exception handling.